Investigating the Tone-Segment Asymmetry in Phonological Counting: Three Artificial Language Learning Experiments

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Abstract

Tone-segment asymmetry has been a frequently discussed topic in linguistic typology where tone has been argued to demonstrate properties that are rarely observed in segmental systems. This study asks whether this typological asymmetry corresponds to a systematic difference in learnability, and thus whether learning biases may contribute to the observed typological distribution. We conducted three artificial language learning experiments examining phonological counting, a non-local phonological process that has been attested for tone but not for segments. The results show that the learning profiles of tonal and segmental counting rules closely mirror the typological distribution: segmental counting patterns, which are typologically rare, are consistently harder to learn. This learning asymmetry holds across different participant populations and generalizes across both consonantal and vowel features, suggesting that it reflects a robust and systematic bias. The findings provide experimental evidence that learning biases may contribute to the typological skew between tonal and segmental systems.

Keywords.

Learnability, tone, tone-segment asymmetry, phonological counting, artificial language learning

1. Introduction

Tone systems have long been argued to have "properties that surpass segmental and metrical systems" (Hyman, 2011). Tone is different in the sense that it can do everything that segmental phonology can do, and more. Such a tone-segment asymmetry has been observed in terms of several phonological phenomena including unbounded circumambient high tone plateauing, where any number of tone-bearing units between two underlying high tones also becomes high, (Stevick, 1969; Hyman, 1998; Yip, 2002; Hyman, 2011), first-last harmony, in which the suffixal tone agrees with the word-initial tone, (Bickmore, 2007; Rolle and Bickmore, 2022; Hyman, 2023) and phonological counting, where tonal processes operate over a non-local span that appears to invoke a counting mechanism (Marlo et al., 2015; Paster, 2019).

Previous work has probed the typological asymmetry between tone and segment from different theoretical perspectives. Within Formal Language Theory studies have characterized segmental and tonal patterns in terms of their respective computational complexities (Heinz and Lai, 2013; Chandlee, 2014; Jardine, 2016). In particular, Jardine (2016) examined circumambient unbounded processes observed in tonal but rarely in segmental phonology, arguing that tonal phonology is computationally more complex, as it is not restricted by weak determinism, i.e., the requirement that well-formedness be computable from bounded, locally available information, whereas segmental phonology is usually weakly deterministic. Based on data from Kuria, Rolle & Lionnet (2019) propose a representational account, couched within the framework of Optimality Theory (OT; Prince & Smolensky, 1993; 2004), that explains tonal counting patterns by using templates. They argue that tone may carry an abstract melodic template, whereby a high tone is associated with a floating moraic tier (i.e., "phantom structure") in a morpheme's lexical entry, which enables surface counting patterns to emerge without actually invoking a high tone assignment process with a counting mechanism. While these approaches suggest that there may be substantive differences underlying the observed asymmetry between tone and segment, it remains unclear whether such differences reflect genuine biases in the phonological system. Alternatively, given the relative rarity of counting patterns of any kind, it is possible that we simply haven't studied enough languages to rule out the possibility that segmental counting patterns also occur. In fact, McCollum et al. (2020) have described a case of ATR harmony in Tutrugbu and related languages in which the harmony rule exhibits unbounded circumambient behavior that had once been believed to only occur in tonal rules, which suggests that segmental phonology may require comparable phonological expressivity. On this view, the apparent typological asymmetry between tone and segment may reflect gaps in the available data, rather than a genuine difference between the two domains.

In this study, we address the debate raised above in the context of phonological counting, non-local phonological processes that are potentially featured by a counting mechanism, and ask whether the apparent limitation of segmental phonology is simply due to structural properties of tone vs. segmental system or there are any additional learning bias that underlies this synchronic typological asymmetry. In other words, we aim to pin down whether the typological asymmetry between tone and segment corresponds to a systematic difference in learnability. Adopting an Artificial Language Learning paradigm, we designed a set of artificial languages whose pluralization rules varied along two dimensions: 1) rule type: whether the rule required counting or not (i.e., counting vs. non-counting) and 2) feature type: whether the rule affected a tone or a segment (i.e., tone vs. segment). This design allows us to examine whether morphophonological rules involving segmental counting are more difficult to learn than their tonal counterparts, as well as rules that do not involve counting.

We conducted two experiments. First, we tested two distinct populations, Cantonese-English bilinguals (Experiment 1a) and English monolinguals (Experiment 1b), to assess whether the observed learning profile is robust and stable across different language backgrounds. Second, with respect to the feature type, we contrasted tone with both consonantal (*nasal* in Experiment 1a & 1b) and vowel features (*vowel backness* Experiment 2), which allows us to evaluate whether the asymmetry is category-level (i.e., tone vs. segment) or feature-specific (i.e., tone vs. nasal/backness). Across both experiments, the results reveal a consistent correspondence between learning and typology: segmental counting patterns, which are typologically rare, are also more difficult to learn. These findings suggest that the tone-segment asymmetry is not illusory caused by limited data or diachronic accidental gap, but instead appears to be grounded in learning biases.

Section 1.1 provides an overview of tone-segment asymmetry in the context of phonological counting. Section 1.2 outlines the hypotheses and predictions of the study. Sections 2-4 present the methods and results of the two experiments (Experiment 1-2). Section 5 discusses the implications of the study and then conclude.

1.1 Tone-segment asymmetry in phonological counting

It has long been observed that phonological systems are subject to strong locality restrictions. As argued by McCarthy and Prince (1986), phonological rules typically operate over strictly local domains, evaluating a given element and its immediately adjacent context, without extending further. Under this view, phonological computation is effectively limited to counting "up to two". This anti-counting restriction is reflected in multiple domains of phonology, including the locality of segmental and suprasegmental processes, as well as the maximally binary[1] structure of metrical feet (Kager, 2012).

At the same time, a growing body of work on Bantu tone systems (e.g., Odden, 2003; Marlo, Mwita & Paster, 2015) has identified phenomena that appear to require reference to more distant

[1] Ternary feet, in this context, do not assume counting to three. Kager (2012) proposes the notion of weakly-layered feet in metrical theory, arguing that feet can be maximally ternary, consisting of a binary head and a monosyllabic adjunct, e.g., σ[σσ] or [σσ]σ.

positions, suggesting that phonological computation may in some cases extend beyond strictly local domains. This raises a broader theoretical question as to whether counting should be permitted in phonological grammars. In this paper, however, I do not take a position in this debate. Instead, I use *phonological counting* (or *counting*) as a descriptive, theory-neutral term to refer to non-local dependencies that appear to involve reference to specific positions within a prosodic domain.

Typologically, tone and segmental processes differ sharply in their interaction with counting. While both domains readily exhibit local, non-counting processes, counting patterns show a clear asymmetry: tonal systems permit counting over multiple positions, whereas segmental processes are rarely, if ever, attested to operate over non-local domains of this kind. In particular, tonal processes may target positions several units away (e.g., third or fourth mora), suggesting a degree of computational flexibility not typically observed in segmental phonology. For instance, in Kuria (E.40), an H-tone is assigned to the initial mora of the verb to mark past tense, but it can also be assigned to the third or fourth mora to mark Remote Future and Inceptive, respectively, as illustrated in (1).

(1) Kuria H-tone assignment (Marlo et al., 2015; Paster, 2019: 48)

a. n-to-o-hɔ́ɔ́tɔ́ɔ́t-ɛ́r-a	Past
FOC-1PL-PAST-reassure-APPL-FV	
‘we reassured’	
b. n-tɔ-ɔka-hoótóót-ééy-e	Past progressive
FOC-1PL-PAST.HAB-reassure-APPL.PERF-FV	
‘we have just been reassuring’	
c. n-to-re-hɔɔtɔ́ɔ́t-ɛ́r-a	Remote future
FOC-1PL-FUT-reassure-APPL-FV	
‘we will reassure’	
d. to-ra-hɔɔtɔɔ́t-ɛ́r-a	Inceptive
1PL-INCEPT-reassure-APPL-FV	
‘we are about to reassure’	

With respect to segmental rules, they are almost exclusively edge-oriented, with no counting patterns involved. For example, in the Gurage (Ethiopian Semitic) language Chaha, palatalization of the last root consonant marks the verbal agreement with a second person feminine singular subject, as shown in (2). This rule fails to apply to those roots where the final consonant is non-palatalizable.

(2) Palatalization as verbal agreement in Chaha (McCarthy, 1983)

2nd.m.sg	2nd.f.sg	
nəkəs	nəkəsʲ	'bite'
nəmæd	nəmædʲ	'love'
fəræx	fəræxʲ	'be patient'

A similar example can be found in Tereno language (Bendor-Samuel, 1960), where 1st person and 3rd person possessive forms are distinguished by nasalization of the word-initial segment, which then undergo spreading up to the first stop or fricative in the word, e.g., ˈowuku 'his house' → ˈõw̃ũŋgu[2] 'my house'. The summary table below (though not exhaustive) illustrates a notable typological asymmetry regarding whether morpho-phonological rules allow the non-local counting pattern of tone and segment. The generalization that emerges is that tonal features can be counted over tone-bearing units, whereas segmental features are rarely, if ever, counted in an analogous way.

	Tone	Segment
Counting	Attested e.g., H-tone assignment to the third/fourth mora to mark inceptive in Kuria (E40; Marlo et al., 2015)	Not attested
Non-counting	Attested e.g., H-tone assignment to the first mora to mark past tense (E40; Marlo et al., 2015)	Attested e.g., Palatalization of root-final consonant in Chaha verbal agreement (McCarthy, 1983)

Table 1. Typological Profile of Counting in Tonal vs. Segmental (Morpho-)Phonology

1.2 Hypotheses and predictions

[2] During the spreading, there is an accompanying replacement process targeting the first stop/fricative in the word, where the stop *k* is replaced by a nasal consonant cluster *ŋg*.

This typological pattern is consistent with three possible hypotheses. First, the observed asymmetry could reflect the process by which these rules are learned: perhaps it is easier for people to learn tonal counting rules than segmental counting rules. Second, this pattern could arise via the historical pressures that shape languages: perhaps tonal counting rules are more stable over long periods of time than segmental counting rules for reasons that have little to do with individual learning processes. Finally, this apparent asymmetry could reflect accidental gaps in the existing data. The total number of attested tonal counting rules is relatively small. If counting rules are in general quite rare, then we would expect the number of segmental counting rules to be small as well. Thus, it is possible that they do exist, but we have failed to find them, or that they are not present in the world's languages due to random variation in the languages that are created and happen to survive. The present study explores the first hypothesis by assessing whether learning patterns are consistent with the typological profile that we observe. Specifically, we ask whether it is easier for people to learn tonal counting rules than segmental counting rules.

In these experiments, we manipulate both feature type (tone vs. segment) and rule type (counting vs. non-counting) in a factorial design. This design yields four types of pluralization rules. In the tonal non-counting condition, the plural form is realized by assigning a high tone (H) to the first mora of the word, mimicking the Past tense morphology in Kuria. In the segmental non-counting condition, the plural form is realized by assigning a segmental feature to the first consonant or vowel of the word, serving as the segmental counterpart to the tonal rule. In the tonal counting condition, the plural form is marked by assigning an H tone to the third mora of the word, modeled on the Inceptive morphology in Kuria. In the segmental counting condition, the plural form is marked by assigning a segmental feature to the third consonant or vowel of the word, representing a counting version of the segmental rule. Each participant was randomly assigned to learn one of the rules and then tested on new examples of that rule to assess what they had learned.

Figure 1 illustrates three of the possible data patterns that we might see. If counting rules in general are hard to learn, but segmental rules are no harder than tonal rules, then we would expect the pattern shown in panel A. This pattern would be broadly compatible with our second and third hypotheses about the typological pattern, suggesting that the asymmetry is either an illusion or the result of historical processes that is not accompanied by any synchronic learning bias. The signature of this data pattern is a main effect of counting/non-counting (with performance on counting being lower), no effect of segment/tone, and no interaction between these variables. The second possibility (panel B) is that there are two independent sources of difficulty: rules over segments are harder to learn than rules over tones and counting rules are harder to learn than non-counting rules. On this hypothesis, we might expect counting rules over segments to be rare even if there was no particular difficulty in learning rules that have both features. The statistical signature of this pattern is a main effect of counting/non-counting (with performance on counting being lower), a main effect of segment/tone (with segment being harder), and no interaction between these variables. Finally, and most critically, it is possible that rules which involve counting over segments are uniquely challenging to learn (panel C). This pattern would support the hypothesis that there is a systematic asymmetry in the way in which we learn phonological rules that gives rise to the observed typological asymmetry. The statistical signature of this pattern is a main effect of counting/non-counting (with performance on counting being lower), a main effect of segment/tone (with segment being harder), and an interaction between rule type (counting/non-counting) and feature type (segment/tone), such that the advantage for tone over segment is more pronounced in the counting condition. If the typological absence of segmental counting rules is attributable solely to historical processes that are unrelated to learning or to random chance, there would be no reason to expect an interaction of this kind.

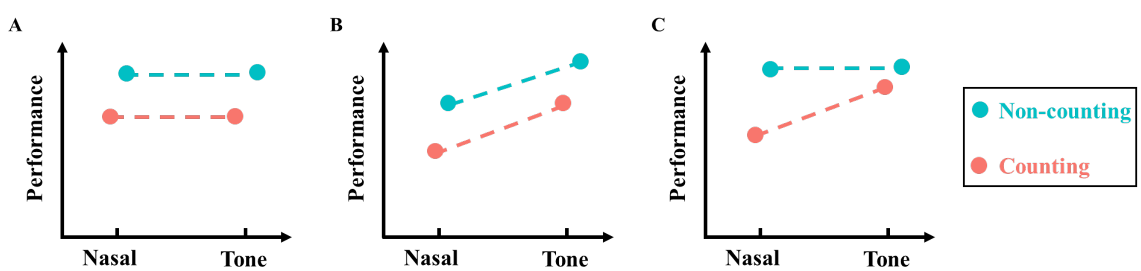

Figure 1.

Three possible data patterns: one in which counting rules are harder to learn than non-counting rules (A), one in which rule type and feature type have independent effects (B), and one in which learners have a particular difficulty with segmental counting rules (C).

2. Experiment 1a

2.1 Methods

All experiments reported in this paper were approved by the Harvard University Institutional Review Board.

2.1.1 Participants

80 Cantonese-English bilingual adults were recruited through Prolific (N = 8) or through personal networks and outreach on The University of Hong Kong campus (N = 72). Each condition had 20 participants, with 2 recruited from Prolific and 18 recruited offline. All participants were at least 18 years old and were native Cantonese speakers who were conversationally fluent in English. Our sample size (N=80; 20 per condition) was chosen based on previous, similar, experimental phonology studies (e.g., Morley, 2018). Participants gave informed consent and received $7.5 USD (or $25 HKD) for completing the study.

Cantonese speakers were chosen as the target population because the tonal stimuli designed for the experiment are level tones from the Cantonese tonal inventory, which they can easily perceive. Participants were required to have good knowledge of English to ensure that they were familiar with multisyllabic words since the words used in the experiment had 3-5 syllables.

Participants were excluded from the analysis if they failed the attention check or did not complete the experiment. All participants passed the attention check and completed the study and were therefore retained. Participants who achieved less than 80% accuracy in Test 1 were required to repeat Test 1 before proceeding and all of them passed on the second try. Departing from our preregistered plan, we retained participants who repeated Test 1, as excluding them would have resulted in substantial data loss. The preregistered analysis is reported in the Supplementary Materials. Thus, the final sample consisted of 80 participants.

2.1.2 Procedure

Participants completed a forced-choice task administered through using the online experiment platform PCIbex (Zehr & Schwarz, 2018). Participants were told to learn how to realize the plural form of a known object in an artificial language. The experimental paradigm was composed of two phases which were each made up of a training and a test component. In the training components, following Morley's paradigm (2018), participants were trained on a plural rule in the language involving a particular morpho-phonological change (in tone or nasal). In the test components they were tested on their ability to generalize the learned rule to new singular-plural pairs of objects.

Phase One

The training component of Phase 1 (Training 1) consisted of 9 trials. Each training trial consisted of two audio descriptions of the images, one corresponding to the singular instance of the object and the other to the plural instance. On each trial, participants first saw the singular object accompanied by the audio label. After playing the audio label, the image of the plural object appeared accompanied by a corresponding audio label. After playing this second audio, participants were prompted to continue to the next trial. Participants were given the option of replaying each audio and no limit was placed on the number of plays. This process was repeated for each training trial. A sample training trial is shown in Figure 2.

Figure 2

Example of a training trial from the nasal-counting condition.

No written words appeared on the screen. The text in italics represents what participants heard when playing each sound file.

In the test component of Phase 1 (Test 1) consisted of 9 trials. On each trial, participants first saw the singular object accompanied by the audio label. After playing the audio label the image of the plural object appeared accompanied by a corresponding audio label. After playing this first possible plural audio, a second possible plural audio appeared. Finally, after this file was played, participants were prompted to choose between the two. One of these choices was the correct plural form according to the rule demonstrated during the training phrase, while the other choice (the foil) was either identical to the singular form or it included a morphological change not present in the training. Specifically, the correct form (i.e., the trained change: high tone 55 or nasalization) was contrasted with a foil in which the trained change was replaced by an unattested alternation that never appeared during training. For example, in the tone conditions, instead of a high-level tone (55) on the first/third syllable, the foil form contained a high rising tone (24). In the segmental conditions, instead of nasalization on the first/third consonant, the foil form exhibited a [+spread glottis] feature.

The order of appearance of the audios (correct and incorrect plural forms) was based on a fixed list that was randomly created. Participants were given the option of replaying audio labels and no limit was placed on the number of plays. This process was repeated for every test trial. An example Test 1 trial is illustrated in Figure 3.

Figure 3

Example of test trial from Phase 1 (Test 1) in the nasal-counting condition.

No written words appeared on the screen. The text in italics represents what participants heard when playing each sound file.

We expected that, if participants were attending to the task, they would perform well in this test phase since three of the foils involved no change from the singular form and six involved a novel phonological change that had not been heard in training. These changes could be detected without correctly learning a counting or non-counting rule, since the correct response could be chosen without tracking the position of the change. Thus, participants were only allowed to *directly* proceed to Phase Two if they were at or above 80% accuracy in Test 1. If participants were below this threshold, they were automatically returned to the beginning of Test 1 and given a second chance to pass it.

Phase Two

Phase 2 began with a training block (Training 2) with the same structure as the training block in Phase 1, but with different lexical items. The test component of Phase 2 (Test 2) consisted of 18 trials which had the same trial structure as Phase 1 except for the audio foils. Specifically, the correct plural audio was contrasted with an audio in which the morpho-phonological change that appeared during training was placed in the wrong position within the word. To successfully distinguish these foils from the target form, participants would have to have learned the correct rule for the position of the morpho-phonological change.

After completing Phase 2, participants completed a brief-post task questionnaire about their linguistic and musical background. The full experiment session took approximately 20-30 minutes.

2.1.3 Materials

Experiment 1a used a 2 × 2 between-subjects design with four conditions. The two factors were feature type (tone vs. nasal) and rule type (counting vs. non-counting). For instance, for participants who were assigned to a tone counting condition, they will learn to assign a H-tone to the third mora to realize the plural morpheme. A summary of the four conditions is provided in Table 2.

	Tone	Nasal
Counting	H-assignment to the third mora	[+nasal]-assignment to the third consonant
Non-counting	H-assignment to the first mora	[+nasal]-assignment to the first consonant

Table 2. Summary of Experimental Conditions: Feature Type × Rule Type

There were 45 pairs of singular-plural object images and each participant saw a full set of them. The images of the objects were sourced from unsplash.com and no objects were repeated across the experiment. The audio stimuli were created by pseudo-randomly combining segments from a subset of the Cantonese inventory (oral stops: /p, t, k/; corresponding nasals: /m, n, ŋ/; corresponding aspirated stops: /p^h, t^h, k^h/, vowels: /i, ɛ, u, ɔ, a/). Tonal stimuli involved include a high-level tone (55), mid-level tone (33) and a high rising tone (24). All the words in the stimulus set consist only of syllables without codas, and none of the multisyllabic stems are real words in Cantonese.

During the two training blocks and Test 1, participants were presented with nine nonce items: three trisyllabic, three quadrisyllabic, and three pentasyllabic forms. Test 2 included 18 nonce items, with six items at each syllable length. All singular forms bore a mid-level tone (33). The realization of the plural marker differed across conditions. In the tone/nasal-counting conditions, plural marking was implemented by placing either a high-level tone (55) or a [+nasal] feature on the third mora or consonant of the word. In the corresponding non-counting conditions, the same feature was instead realized on the initial mora or consonant.

Incorrect plural forms were designed differently for two tests. In Test 1, across all conditions, three of the incorrect forms were singular forms with no phonological alternation, while six were hypothetical plural forms that involved an untrained phonological change applied to the correct position in the word. In the tonal conditions, the incorrect phonological change was a high rising tone (24) rather than the expected high-level tone (55). In the nasal conditions, the incorrect change involved the feature [+spread glottis] instead of the expected [+nasal]. In Test 2, incorrect plural forms were created by assigning the target high-level tone or [+nasal] feature to the wrong position, specifically to the second mora or consonant rather than the intended location.

In total, we created 45 singular-formed words, 45 × 4 correct plural-formed words, 6 × 4 incorrect plural-formed words with wrong featural change and 18 × 4 incorrect plural-formed words with wrong positional change. All audio labels were recorded by a native Cantonese speaker and were recorded in one session in a sound-proof booth at the MIT Phonetics Lab. All materials (images and audios) are available on our OSF repository.

2.2 Results

2.2.1 Test 1

Unlike Test 2, which specifically assessed participants' ability to learn counting-based rules, performance in Test 1 primarily depends on participants' ability to perceive the relevant tonal and segmental contrasts. To determine whether this ability to perceive the contrast varied across conditions, we analyzed participants' overall performance on Test 1.[3] A logistic mixed-effects regression model (Correct ~ feature * rule + (1|ID) + (1|Trial), with contrast coding) was used to examine the effects of feature type and rule type on the binary outcome variable Correct (Yes/No), as outlined above. The results, plotted in Figure 4 below, revealed no significant main effect of rule type ($\beta = 0.04$, SE = 0.19, z = 0.21, p = .835) or feature type ($\beta = 0.09$, SE = 0.20, z = 0.43, p = .666). There was also no significant interaction between counting and feature ($\beta = -0.23$, SE = 0.19, z = -1.19, p = .236). Thus, performance did not differ reliably as a function of either rule or feature type, nor was there evidence that the effect of feature differed across different counting conditions, suggesting no detectable difference in Cantonese speakers' sensitivity to tonal and segmental contrasts.

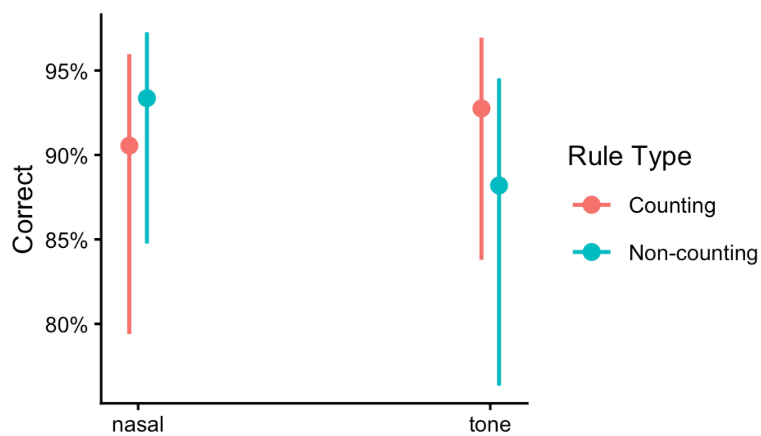

Figure 4 Performance on Test 1 in Experiment 1a.

Test 1 assesses whether learners know the kind of phonological change that marks plurals. Model-predicted probabilities of correct responses in Test 1 from a logistic mixed-effects model including feature type, rule type, and their interaction. Error bars represent 95% confidence intervals.

2.2.2 Test 2

We now analyze the 18 test trials in Test 2, which constitute the critical trials in our design. Test 2 assesses whether participants have learned the position of the relevant phonological rule and thus it allows us to assess whether they have succeeded in learning the critical rule in both the counting and non-counting conditions. As we will see, the findings demonstrate that

[3] A coarser analysis was proposed in the preregistration plan. The conclusions of that analysis are broadly similar to the one reported here.

participants perform very well in every condition except for the segmental counting conditioning, suggesting that the tone-segment asymmetry is attributable to the difficulty of learning segmental counting rules.

Figure 5 presents the Cantonese-English bilingual participants' performance in Test 2. As we can see, accuracy is considerably lower in the counting-nasal condition (63%) than in the non-counting-nasal condition (87%). In contrast, performance in the two tonal conditions is near ceiling, with no difference between counting (92%) and non-counting (93%) conditions. This pattern reveals a clear asymmetry: the effect of counting is present for the nasal conditions but not for the tonal ones.

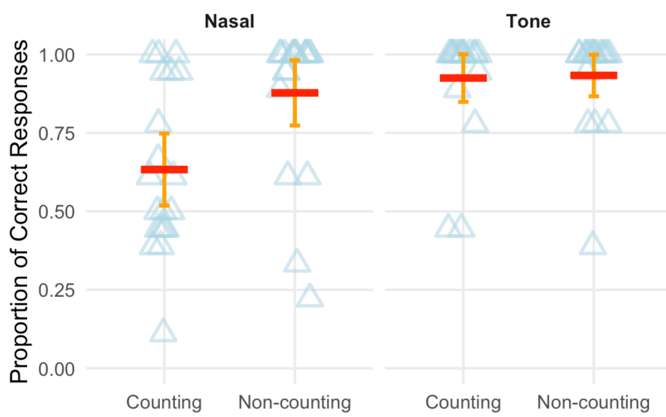

Figure 5 Proportion of correct responses in Test 2 (Experiment 1a) by feature type and rule type. Test 2 assesses whether learners know the position of the phonological change that marks plurals. Each point represents the proportion correct for an individual participant, the red horizontal lines indicate group means for each condition. Error bars represent 95% confidence intervals.

To statistically evaluate the patterns outlined above, we fit a logistic mixed-effects model, implemented in R (Bates et al., 2015). The binary dependent variable (DV) *Correct* encodes the trial level accuracy, which is predicted by two independent variables (IV) feature type, rule type, and their interaction. We also include random intercepts for both participants and trials in the model. Thus our final model was specified as *Correct ~ feature * rule + (1|ID) + (1|Trial)*, with contrast coding.[4] The model showed a significant main effect of rule type ($\beta = -1.46$, SE = 0.36, $z = -4.05$, $p < .001$), indicating lower overall accuracy in the counting condition, and a significant main effect of feature ($\beta = -0.83$, SE = 0.32, $z = -2.61$, $p = .009$), showing lower accuracy for nasal than for tone. Crucially, there was also a significant interaction between feature type and rule type ($\beta = -0.78$, SE = 0.39, $z = -2.00$, $p = .045$), indicating that the negative effect of counting is greater for nasal rules. In line with the pattern visible in the effects plot (Figure 6), counting imposes an increased cost on nasal learning but has little effect on tonal performance. This interaction demonstrates that the typologically uncommon nasal counting pattern is more difficult to learn than its typologically more robust tonal counterpart.

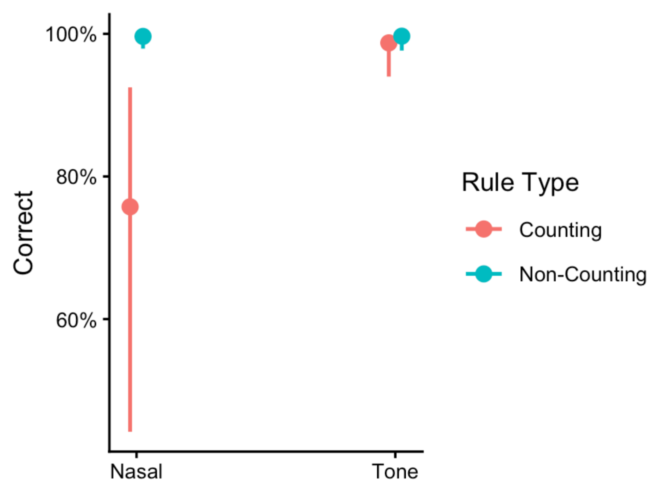

Figure 6 Performance on Test 2 in Experiment 1a.

Test 2 assesses whether learners know the position of the phonological change that marks plurals. Model-predicted probabilities of correct responses from a logistic mixed-effects model. Error bars represent 95% confidence intervals.

In summary, Experiment 1a had two primary findings. First, in Test 1, Cantonese speakers showed broadly similar sensitivity to tonal and segmental features, with no significant difference in performance between the two feature types. Second, in Test 2, the disruptive effect of counting was present when learning a nasal rule but not when learning a tonal rule. These results

[4] Analyses followed the preregistered plan, in which models were built incrementally from a base model to one including the interaction between feature and rule types. For brevity, we report results from the final model here.

are clearly compatible with the learning hypothesis: the pattern observed in Test 2 is precisely what we would expect if the typological skew between non-local tonal and segmental processes was attributable to a learning bias. These results, however, are also compatible with an alternative interpretation: perhaps the tonal contrast in the present study is so easy for Cantonese speakers to detect that they have ample bandwidth left to learn the more difficult counting rule. On this hypothesis, the added difficulty of learning a counting rule might be similar between segmental and tonal contrasts, but this would be undetectable in this population because of ceiling effects in the tonal conditions. This alternative interpretation predicts that the asymmetry should disappear if we test participants for whom tonal contrasts are not as salient. To explore this, we tested we tested English speakers with no tone language experience in Experiment 1b.

3 Experiment 1b

3.1 Methods

Experiment 1b differed from Experiment 1a only in the language background of the participant group; all other aspects of the materials, procedure, and stimuli were held constant.

3.1.1 Participants

A total of 120 monolingual English-speaking adults were recruited via Prolific. Assignment to experimental conditions was determined by the platform and resulted in unequal sample sizes across conditions. To ensure equal numbers of participants in each condition, only the first 20 participants who completed the study in each condition were included in the analysis. The final sample therefore consisted of 80 participants. All participants were at least 18 years old and self-identified as native monolingual speakers of English. All participants provided informed consent and received USD 7.50 for their participation.

3.1.2 Procedure

The procedure was identical to that of Experiment 1a.

3.1.3 Materials

The materials were the same as in Experiment 1a.

3.2 Results

The goal of Experiment 1b was to determine whether counting would continue to have a larger impact on the learnability of segmental rules in a population where tonal contrasts were harder to detect.

3.2.1 Test 1

We analyzed participants' overall accuracy in Test 1 as an index of learning difficulty across conditions. A logistic regression model (Correct ~ feature * rule + (1|ID) + (1|Trial), with contrast coding) shows a significant main effect of feature ($\beta = 1.28$, SE = 0.34, $z = 3.73$, $p < .001$), indicating English monolinguals' performed better in the nasal conditions than in the tonal conditions. While there is no main effect of rule type ($\beta = -0.05$, SE = 0.31, $z = -0.17$, $p = .863$), a significant interaction between feature and rule types is observed ($\beta = -1.14$, SE = 0.32, $z = -3.51$, $p < .001$), which is potentially driven by the particularly low accuracy in the non-counting tone condition.[5] The effects plot in Figure 7 below illustrates these effects.

Critically, this pattern is markedly different from that observed in Experiment 1a (cf. Figure 4). Whereas Cantonese speakers showed comparable sensitivity to changes in tone and in segments, English speakers found it easier to detect changes in segments. Thus, if counting rules are merely harder to learn when a contrast is less salient, then, in Test 2, English speakers should acquire the segmental counting rule easily but struggle with the tonal counting rule. In contrast, if there is a stable tone-segment learning asymmetry for non-local rules, then English speakers should pattern with Cantonese speakers in Test 2.

[5] One possible account behind such low performance in tonal non-counting than in tonal counting may be grounded in English speakers' experience with boundary tones in their native language, which may facilitate the perception of tonal contrasts at the right edge of the word.

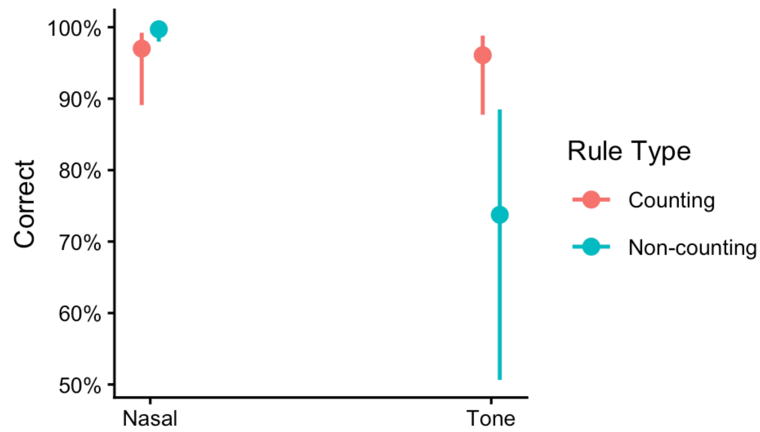

Figure 7 Performance on Test 1 in Experiment 1b.

Test 1 assesses whether learners know the kind of the phonological change that marks plurals. Model-predicted probabilities of correct responses in Test 1 from a logistic mixed-effects model including feature type, rule type, and their interaction. Error bars represent 95% confidence intervals.

3.2.2 Test 2

Figure 8 displays the performance of English monolingual participants in Test 2. The pattern is similar in several ways to one observed in Cantonese learners. In the nasal conditions, there is a sharp drop in performance between the non-counting condition where performance is nearly perfect (95%) to the counting condition where it is barely above chance (56%). In contrast, in the tonal conditions, there is little to no difference between counting (75%) and non-counting (81%) conditions. There is however, one critical difference between the two studies: English speakers show lower overall performance for the tonal conditions (ruling out the possibility of ceiling effects).

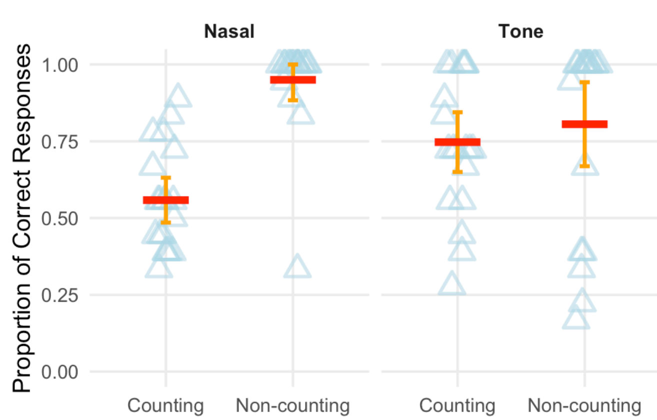

Figure 8 Proportion of correct responses in Test 2 (Experiment 1b) by feature type and rule type.
Test 2 assesses whether learners know the position of the phonological change that marks plurals. Each point represents an individual participant's accuracy, the red horizontal lines indicate group means for each condition. Error bars represent 95% confidence intervals.

The pattern observed in Experiment 1b is supported by the statistical model. Using the same logistic mixed-effects model as in Experiment 1a, we find a robust main effect of rule type (β = -1.50, SE = 0.29, z = -5.10, p < .001). Crucially, this effect interacts with feature (β = -0.85, SE = 0.27, z = -3.11, p = .002), indicating that the drop in performance for counting rules is greater for nasal rules than for tone rules. Unlike Experiment 1a, there is no overall difference between feature types (β = 0.12, SE = 0.27, z = 0.45, p = .654), suggesting that the asymmetry does not arise from a general advantage for a specific feature type but rather from how counting affects the acquisition of rules over the two feature types. This pattern is illustrated in the effects plot (Figure 9). We also conducted the planned pairwise comparisons. Performance did not differ reliably between the counting and non-counting conditions for tone (OR = 0.27, z = - 1.74, p = .304), whereas performance was significantly lower in the counting than in the non-counting condition for nasal (OR = 0.009, z = - 5.52, p < .001).

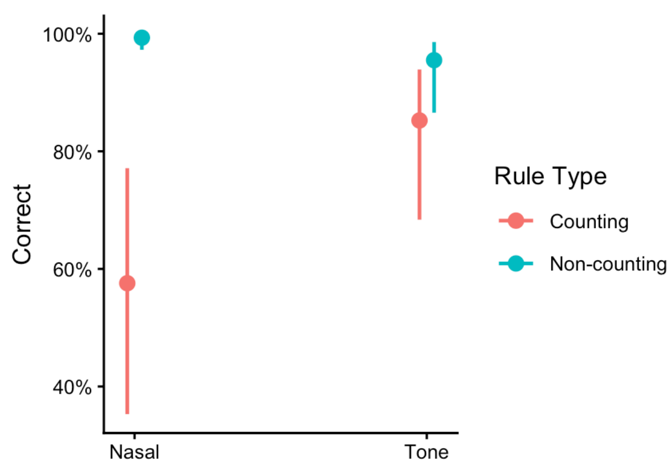

Figure 9 Performance on Test 2 in Experiment 1b.
Test 2 assesses whether learners know the position of the phonological change that marks plurals. Model-predicted probabilities of correct responses from a logistic mixed-effects model. Error bars represent 95% confidence intervals.

In short, the pattern of results in Experiment 1b replicated the critical interaction in rule learning and ruled out an alternative explanation based on contrast salience. Specifically, the results for Test 1 demonstrated that English speakers, as expected, found it easier to learn the relevant perceptual contrast for the nasal rules than for the tonal rules, reversing the pattern observed in the Cantonese speakers. Despite this perceptual disadvantage for tones, English speakers more readily acquired a counting rule based on tone than one based on nasal segments (Test 2). Thus the observed bias favoring tonal over segmental counting rules cannot be attributed to the perceptual salience of a specific feature type as modulated by participants' native language background. Instead, the advantage for tonal counting rules appears to reflect a general learning bias that is strong enough to be visible even in a population that struggles to detect whether the correct tone is present. The difference in effects is striking: for nasal rules performance drops from ceiling to floor when counting is required, for tonal rules it barely budges.

The findings of Experiments 1a and 1b clearly support the learnability hypothesis for the tone-segment asymmetry in counting rules. So far, however, we have tested just one kind of segmental rule; the asymmetry hypothesis predicts that learners should struggle to learn counting rules across the full range of segment contrasts. Experiment 2 tests this prediction by extending our paradigm to a rule in which plurality is marked by vowel backness.

4. Experiment 2

4.1 Methods

4.1.1 Participants

A total of 178 monolingual English-speaking adults were recruited via Prolific. All participants were at least 18 years old and self-identified as native monolingual speakers of English. Four participants failed the attention check and were therefore excluded from the final analysis. The final dataset consisted of 174 participants. Participants were distributed across conditions as follows: Tone-Counting (N = 28), Nasal-Counting (N = 21), Vowel-Counting (N = 33), Tone-Non-counting (N = 29), Nasal-Non-counting (N = 32), and Vowel-Non-counting (N = 31).[6] All participants provided informed consent and received USD 7.50 for their participation.

[6] In the Experiment 1b with a 2 × 2 design, enforcing equal cell sizes (N = 20 per condition) required excluding a relatively modest proportion of the collected data. In contrast, in the Experiment 2 with a 2 × 3 design, enforcing

4.1.2 Procedure

The procedure was identical to that of Experiments 1a and 1b, except that participants were neither required nor able to redo Test 1 due to technical issues.

4.1.3 Materials

Experiment 2 differed from Experiment 1a and 1b in one critical way: we added two vowel conditions to the original design, leading to a 2 × 3 between-subjects design with six conditions. The two factors were feature type (tone vs. nasal vs. vowel) and rule type (counting vs. non-counting). A summary of the six conditions is provided in Table 3.

	Tone	Nasal	Vowel
Counting	H-assignment to the third mora	[+nasal]-assignment to the third consonant	[+back]-assignment to the third vowel
Non-counting	H-assignment to the first mora	[+nasal]-assignment to the first consonant	[+back]-assignment to the first vowel

Table 3. Summary of Experimental Conditions: Feature and Counting Factors

The visual stimuli, were identical to those used in the prior experiment, as were the auditory stimuli for the tone and nasal conditions. In Experiment 2, the trained vowel alternation targeted the front-back distinction. The stimuli for this vowel shift condition were constructed as follows.

The singular items used in training and at test were identical to those used in the previous studies (and in the other conditions). The correct plural form for the new vowel condition, which was used in training and test, was constructed by adding a [+back] feature to the third vowel in the counting condition and to the initial vowel in the non-counting condition.

As in the previous study, incorrect plural forms were constructed differently in Test 1 and Test 2. In Test 1, three incorrect options consisted of unmodified singular forms (i.e., no phonological alternation), while six involved an untrained (and thus incorrect) phonological alternation applied at the correct position in the word. The untrained alternations for the vowel conditions were constructed by replacing the feature [+back] with the feature [+low]. In Test 2,

equal cell sizes would have required trimming the dataset to the size of the smallest cell, resulting in substantial data loss. We therefore retained all eligible participants in the 2 × 3 experiment.

incorrect plural forms were created by assigning the correct plural feature, [+back], to the wrong word position, namely the second vowel rather than the intended target location (i.e., the first or third vowel). All stimuli consisted of open syllables, and none of the multisyllabic stems corresponded to real English words.

To ensure that the stimuli in all conditions were comparable, we recorded all the audio labels used in Experiment 2 with a single Cantonese speaker in one session in a sound-proof booth at the MIT Phonetics Lab. All materials (images and audios) are available on our OSF repository.

4.2 Results

4.2.1 Test 1

We built up the statistical model incrementally, beginning with a model containing counting as a fixed effect. We then added feature as a fixed effect, followed by the interaction between counting and feature (i.e., culminating at the full factorial model (Correct ~ feature * rule + (1|ID) + (1|Trial)). At each step, we compared the more complex model with the preceding model using likelihood-ratio tests. We found no main effect for rule type (β = - 0.35, SE = 0.23, z = - 1.53, p = .126). Adding feature type significantly improved the model fit, $\chi^2(2) = 17.50$, p < .001, indicating an overall effect of feature type. Crucially, including the interaction between rule and feature type further improved model fit, $\chi^2(2) = 23.85$, p < .001. Specifically, for tone stimuli, performance in the non-counting condition is much lower than performance in the counting condition, much as it was in Experiment 1b. This suggests that English speakers have particular difficulty identifying the correct tone in initial position. In contrast, performance in the counting and non-counting versions of both the nasal and vowel conditions is similar resulting in interactions for both kind of segmental stimuli relative to the tone conditions (nasal (β = 2.90, SE = 0.61, z = 4.77, p < .001) and vowel (β = 1.27, SE = 0.51, z = 2.46, p = .013). The effects plot of the model is provided in Figure 10.

To further unpack the interaction terms, we conducted separate analyses for counting and non-counting conditions. In the counting condition, performance for the nasal and tonal conditions did not differ significantly (β = - 0.55, SE = 0.45, z = - 1.22, p = .223), and likewise, there was no significant difference between the vowel and tonal conditions (β = - 0.79, SE = 0.41, z = - 1.91, p = .056). In the non-counting condition, however, performance was significantly higher in the nasal than in the tonal condition (β = 2.51, SE = 0.44, z = 5.74, p

< .001), whereas performance did not differ significantly between the vowel and tonal conditions ($\beta = 0.53$, $SE = 0.33$, $z = 1.59$, $p = .112$). To directly compare nasal vs. vowel, we also conducted a pairwise analysis. The results showed that there is no significant difference between nasal and vowel in the counting condition ($OR = 1.24$, $z = 0.55$, Tukey-adjusted $p = .849$), whereas performance in nasal condition was significantly higher than that in vowel condition under non-counting ($OR = 6.34$, $z = 4.44$, Tukey-adjusted $p < .001$). Thus, whereas feature type had little detectable effect under counting, performance on nasal stood out from both tone and vowel under non-counting.

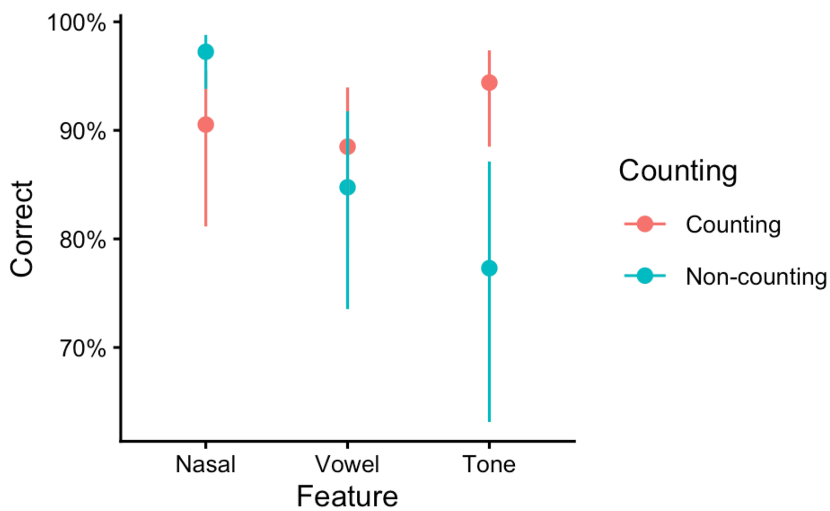

Figure 10 Performance on Test 1 in Experiment 2.

Test 1 assesses whether learners know the kind of the phonological change that marks plurals. Model-predicted probabilities of correct responses in Test 1 from a logistic mixed-effects model including feature type, rule type, and their interaction. Error bars represent 95% confidence intervals.

Two conclusions emerge from this analysis. First, across studies there appears to be modest stability within a population in this measure of how readily the kind of change is identified. In both Experiment 1b and Experiment 2, English speakers struggled in the tone non-counting condition relative to the tone counting condition and the nasal conditions. However, performance for the nasal counting condition was poorer than in the previous study and performance for the tonal counting condition was better. Second, to the extent that performance on Test 1 can be considered a perceptual baseline for Test 2, these results suggest that purely perceptual account would predict worse performance in the tone non-counting condition than in all others.

4.2.2 Test 2

As previewed, Experiment 2 extends the design of Experiments 1a and 1b by introducing vowel backness as an additional segmental feature, providing a critical test of whether the observed asymmetry is feature-specific or reflects a more general category-level distinction between tone and segmental features.

Figure 11 presents participants' performance across the three feature types (tone, nasal, and vowel) under counting and non-counting conditions. In the counting condition, accuracy is highest for tone (75%) and substantially lower for both nasal (60%) and vowel (60%). In the non-counting condition, performance is relatively high across all features, though nasal reaches near-ceiling accuracy (98%), while tone (88%) and vowel (86%) remain slightly lower. This suggests that nasal and vowel behave similarly, in contrast to tone, under counting, whereas in the non-counting condition vowel patterns more closely with tone. While tone remains relatively stable across different rule types, both segmental features (vowel and nasal) show a marked improvement from counting to non-counting, which suggests their greater sensitivity to the demands of counting.

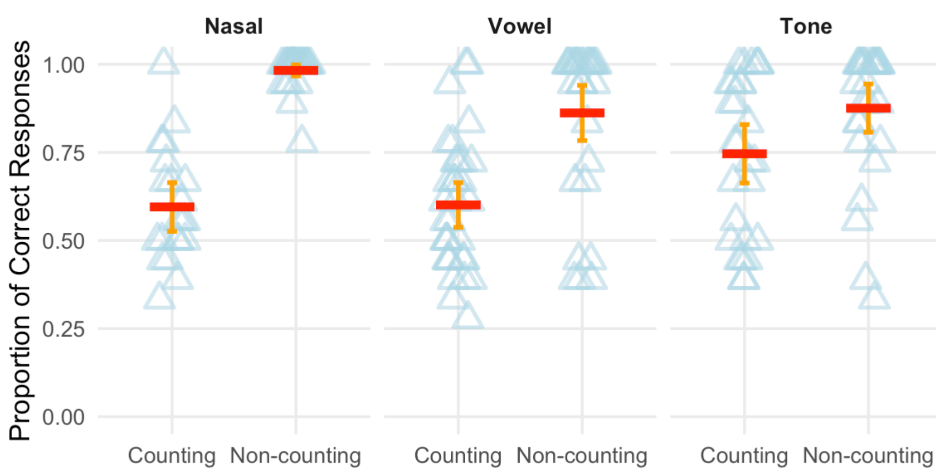

Figure 11 Proportion of correct responses in Test 2 (Experiment 2) by feature type and rule type. Test 2 assesses whether learners know the position of the phonological change that marks plurals. Each point represents an individual participant's accuracy, the red horizontal lines indicate group means for each condition. Error bars represent 95% confidence intervals.

Models were built incrementally by first including rule type as a fixed effect, then adding feature, and finally adding the interaction between rule type and feature. Successive models were compared using likelihood-ratio tests. The initial model revealed a significant effect of rule type, with participants performing better in the non-counting than in the counting condition ($\beta = 2.69$, $SE = 0.32$, $z = 8.51$, $p < .001$). Adding feature did not significantly improve the model fit, $\chi^2(2) = 5.85$, $p = .054$. However, adding the interaction between counting and feature did, $\chi^2(2) = 20.39$, $p < .001$. To explore this interaction, we conducted separate analyses for the counting and non-counting conditions. In the counting condition, performance was significantly lower for nasal than for tone ($\beta = -0.92$, $SE = 0.31$, $z = -2.92$, $p = .003$) and significantly lower for vowel than for tone ($\beta = -0.88$, $SE = 0.28$, $z = -3.11$, $p = .002$). In the non-counting condition, performance was significantly higher for nasal than for tone ($\beta = 2.60$, $SE = 0.87$, $z = 2.99$, $p = .003$), whereas vowel did not differ significantly from tone ($\beta = 0.21$, $SE = 0.80$, $z = 0.26$, $p = .796$). Additional pairwise comparisons showed that nasal and vowel did not differ in the counting condition ($OR = 0.95$, $z = -0.13$, Tukey-adjusted $p = .991$), whereas nasal performance was significantly higher than vowel performance in the non-counting condition ($OR = 8.70$, $z = 3.79$, Tukey-adjusted $p < .001$). These results demonstrate that the asymmetry observed in Experiments 1a and 1b is not specific to nasal vs. tone but extend to vowel backness as well. The effects plot based on the full factorial model is shown in Figure 12.

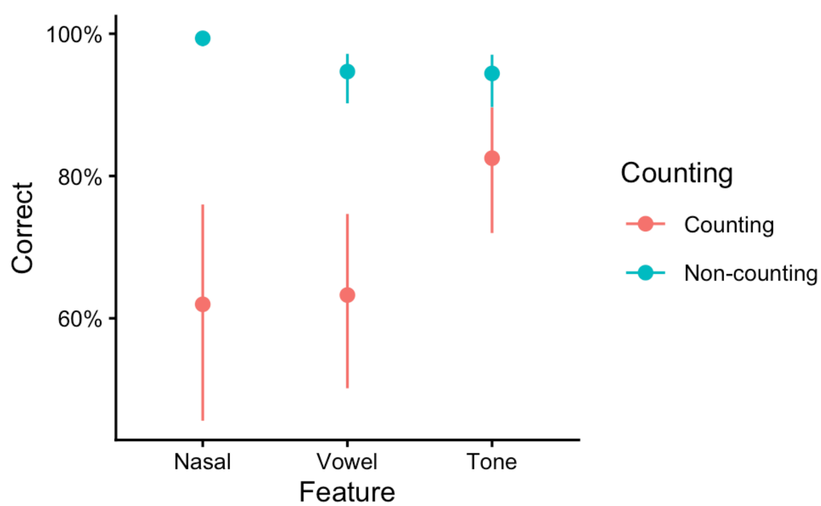

Figure 12 Performance on Test 2 in Experiment 2.

Test 2 assesses whether learners know the position of the phonological change that marks plurals.

Model-predicted probabilities of correct responses from a logistic mixed-effects model. Error bars represent 95% confidence intervals.

Together, these results suggest that the asymmetry observed in Experiments 1a and 1b extends beyond the consonantal nasal feature vs. tonal feature, demonstrating that the learning asymmetry in non-local counting-based phonological rules reflects a category-level distinction rather than a feature-specific effect. In addition, Experiment 2 reveals a more nuanced pattern in the behavior of vowel changes. In Test 1, which primarily taps perceptual encoding, we found that in non-counting conditions the vowel change patterned with the tone change, rather than the consonant change. This pattern persists in the Test 2 non-counting condition. This suggests that vowels behave similarly to tone in phonetic perception, when they are in first position, consistent with their role as primary tone-bearing units.[7] In contrast, in the counting condition of Test 2, which directly assesses the learning of counting-based phonological rules, the vowel condition shows significantly lower performance, comparable to nasal and distinct from tone. This dissociation indicates that, while vowels tend to align with tone perceptually, they pattern with consonantal features during the learning and phonologization of counting-based non-local rules, reflecting their shared segmental status.

5. Discussion and Conclusion

So far, we have obtained a consistent and robust asymmetry in the learning of non-local counting-based rules. In Test 2 across both experiments, tonal counting patterns were consistently more readily acquired than their segmental counterparts. This asymmetry was present in both Cantonese-English bilinguals (Experiment 1a) and English monolinguals (Experiment 1b), demonstrating that the asymmetrical learning profile cannot be reduced to a language-specific advantage for tone. Experiment 2 further showed that the asymmetry is not limited to nasality: vowel backness patterned with nasality rather than tone for the counting rule, suggesting that the contrast is better understood as a broader distinction between tonal and segmental features than as a feature-specific effect.

The discussion is organized as follows. First, I argue that our experimental results support a learnability-based account of the typological asymmetry between tonal and segmental counting

[7] Strictly speaking, tone-bearing units refer to phonological timing units such as moras and syllables. In most cases, vowels serve as the primary hosts of tone, insofar as they contribute moraic structure.

patterns: structures that are less robustly attested cross-linguistically are also harder to learn. Second, I examine the possible learning biases underlying the observed asymmetry. I first consider and reject a purely perceptual explanation, focusing on the contrast between Test 1 and Test 2 across Experiments 1a and 1b. I then turn to the results of Experiment 2, showing that the divergent patterning of vowel across Test 1 and Test 2 further supports a phonological account of the asymmetry, grounded in the distinction between tonal and segmental systems rather than in mere perceptual similarity. Finally, I discuss potential substantive biases that favor tone over segment, as well as the prosodic nature and representational properties of tonal features that may underlie their learning advantage.

5.1 Tone-segment asymmetry and learnability

A central implication of the present findings is that the typological asymmetry between tone and segment is unlikely to be illusory, and does not merely reflect a gap in the available data (see McCollum et al., 2020). Instead, the results suggest that this asymmetry may be grounded in learnability.

Under counting conditions, tonal patterns were consistently easier to acquire than segmental ones. This asymmetric learning profile was observed in both tone-speaking and non-tone-speaking populations and generalizes beyond nasality to another segmental feature, vowel backness. These results demonstrate that the tonal advantage is not limited to those who have already learned tonal languages and suggest that it reflects a broader distinction between tonal and segmental features. Crucially, this learning asymmetry neatly aligns with typological observations: non-local counting patterns are frequently attested in the tonal domain, whereas comparable segmental counting rules are rare or unattested. The present study suggests that the difference in learnability may explain the difference in the typological distribution. Edge-based (non-counting) rules and tonal counting rules are relatively easy to learn so they persist across generations. In contrast, segmental counting rules are quite difficult to learn and thus they either fail to emerge or disappear over time when learners fail to acquire them.

This interpretation of the results is consistent with a growing body of literature arguing that synchronic learning biases shape the observed phonological typology (e.g., Moreton, 2008; Moreton & Pater, 2012a,b; Staubs, 2014; Hughto, 2020; O'Hara, 2021). Previous experimental studies have shown that phonological patterns that are more typologically common are more readily learned than those that are typologically rare. The present findings extend this line of research by showing that such biases are not limited to specific phonological patterns, such as vowel harmony (e.g., Finley, 2008; Finley & Badecker, 2010; Finley, 2012; Martin & White, 2021; Yu & Do, 2021, Huang & Do, 2023), consonant harmony (e.g., Koo & Cole, 2006; Conklin et al., 2023), or epenthesis (Morley, 2018), but also apply to the broad asymmetry between tonal and segmental counting rules. Future work could extend the current paradigm to investigate other phenomena, such as first-last harmony and unbounded circumambient spreading, that instantiate this long-standing tone-segment asymmetry in the phonological literature.

5.2 Why tonal counting is easier to learn than segmental counting

The asymmetry that tonal counting patterns are consistently easier to learn than their segmental counterparts raises the question of what underlying learning biases give rise to this profile. I first evaluate and reject explanations that depend solely on the phonetic properties of tone (Section 5.2.1). I then turn to other possible substantive biases that could underlie the observed asymmetry (Section 5.2.2).

5.2.1 The learning asymmetry cannot be fully reduced to perceptual factors

It may be tempting to attribute the advantage of tonal counting to the greater acoustic salience of tone. The present results, however, do not support this account. In Test 1, participants were asked to detect whether pluralization involved a phonological change and, if so, to identify the nature of the change (but not its location). Thus, performance in Test 1 primarily reflects participants' perception and encoding of the trained phonological alternation. The results demonstrate a clear cross-linguistic difference. Cantonese speakers, who have extensive experience with tone, perform equally well on tonal and nasal contrasts (see Figure 4). In contrast, English speakers who do not know a language with tonal contrasts, perform more poorly on tonal contrasts, particularly those in word initial position (see Figure 7). Thus, the relative difficulty of perceiving tonal and segment contrasts can be strongly shaped by participants' linguistic background. Critically, however, perceptual performance does not reliably predict learning outcomes. While English speakers struggled with tonal contrasts, both English and Cantonese speakers were consistently better at learning positional counting rules for tones than for segments. This dissociation suggests that the observed learning asymmetry cannot be simply explained by the acoustic salience of tone or by learners' perceptibility alone.

The behavior of the vowel rules in Experiment 2 provides a critical test case for evaluating a pure phonetically-based account. Previous work has noted that vowels and consonants tend to serve different roles in language processing (Bonatti, 2005; Toro et al., 2008a,b): vowels are more closely associated with grammatical and prosodic information, whereas consonants are more strongly tied to lexical identity. In addition, vowels share important perceptual properties with tones: vowels are essential to moraic/syllabic structure and therefore serve as tone-bearing units. As such, they have comparable perceptual duration. Like tones, vowels involve continuous acoustic dimensions, which makes them perceptually similar in many contexts. These two factors suggests that vowel features may occupy an intermediate position between tone and consonantal features, making them a useful probe for evaluating whether the observed asymmetry reflects a category-level distinction between segmental and tonal features, rather than a narrower generalization (nasal vs. tone) or a more continuous distinction in the space of perceptual contrasts.

Thus, in Experiment 2, we introduced a new condition, vowel backness. We found that the vowel rules did not consistently pattern with either tone or nasal rules. For the edge-based non-counting rule, vowels patterned with tones: performance was strong but below that of the nasal condition. In contrast, for the counting rules, performance in the vowel condition closely paralleled performance in the nasal condition.

This pattern dispels a purely phonetic or perceptual account of the observed learning asymmetry. If perception-related factors were the main driving learning in all conditions, then vowels would be expected to consistently pattern with tone for both rule types. The fact that both consonants and vowels pattern together against tone under counting conditions, despite their distinct roles in encoding different types of information in language processing and their distinct phonetic precursors, suggests that the asymmetry reflects a fundamental distinction between tonal and segmental systems.

5.2.2 What makes tone different?

The discussion so far suggests that the advantage of tonal patterns cannot be fully reduced to purely perceptual factors. Instead, it appears to reflect deeper differences in how tonal and segmental systems are organized and represented in the grammar. Below we consider four possibilities.

First, the asymmetry could reflect the representational properties of tone. With the advent of autosegmental phonology (Goldsmith, 1976; Leben, 1973), tone has been analyzed as operating on an autonomous tier associated with tone-bearing units. This relative independence allows tonal features to extend across multiple units and to form structured patterns over a long-distance prosodic domain more easily. In the context of the present study, this suggests that tonal representations are naturally well-suited to operations that require tracking relations across positions, such as non-local counting rules. In contrast, although segmental features can also be represented autosegmentally by each projecting their own tiers (Clements, 1976), they are more typically understood as embedded within a more complex featural geometry associated with individual segments, which makes them less compatible with operations that require abstraction over multiple positions.

Second, the prosodic nature of tone, as opposed to segmental features, may also play a crucial role in shaping the observed learning asymmetry. The prosodic status of tone is evident in its participation in meter. Meter regulates only prosodic properties such as stress, weight, and tone (Kiparsky, 2020), and may, for example, require the third syllable of a line to be stressed, heavy, or high-toned. Crucially, no meter regulates segmental properties such as nasality, vowel height, or voicing, which would be typologically unnatural. Therefore, tone is intrinsically prosodic in a way that segmental features do not. This distinction is directly relevant to the present task. Although the patterns investigated here may be described as counting-based (e.g., targeting the third syllable), such effects can also be analyzed in terms of hierarchical prosodic structure rather than literal counting. For instance, in Winnebago, main stress is assigned to the third syllable of the word, a pattern that is descriptively count-based but derived from metrical structure (Halle, 1990). From this perspective, the current task may be understood as fundamentally prosodic in nature. If so, its compatibility with tone may stem from the fact that tone itself is a prosodic feature. Learners may therefore be able to recruit prosodic structure when acquiring tonal patterns, whereas comparable segmental patterns lack such structural support. More speculatively, the connection between tone and prosodic structure may extend to domains such as music, which is also prosodically organized. Rhythmic structures involving recurring patterns (e.g., triple rhythms like the waltz) are pervasive in music, and pitch, the straightforward phonetic correlate for tone, is a central component of musical structure. This parallel further suggests that tonal patterns may be particularly well suited to computations involving rhythmic or position-sensitive structure.

Third, tonal features may enjoy an advantage in sequential pattern processing, as tone has long been argued to exhibit a greater syntagmatic nature. Previous work suggests that speakers are better at extracting pitch patterns across syllables (i.e., tonal melodies) than comparable segmental patterns (Hyman, 2018). This may facilitate the tracking of tonal patterns over longer spans, even in cases where local tonal discrimination is relatively weak. This line of reasoning has also been reflected in analyses that appeal to the special representational properties of tone, in particular its melodic structure. As reviewed earlier, Rolle & Lionnet (2019) introduce the notion of phantom structure, where a high tone is pre-associated with a specific tone bearing unit on an abstract tier. Consequently, tone can encode a melodic template that is pre-specified in the lexicon and subsequently aligned with surface positions.

Finally, segmental features may be further constrained by their tighter integration with segmental structure and by the potential for complex feature interactions. In the present study, we examine the learnability of a long-distance segmental feature assignment rule, whose successful application depends on the availability of appropriate segmental hosts at a designated position. However, such operations may pose more of a challenge for segments than for tones. For instance, not all segments are equally suitable hosts for a given feature (i.e., we may frequently encounter undesirable featural interactions due to featural incompatibility), whereas tonal features are less constrained by such host-related restrictions, due to 1) the smaller tonal featural inventory and 2) fewer problematic tonal interactions resulting from the underspecification of low tones. This relative flexibility may make tonal features more compatible with counting-based phonological computation. Although the present study controls for such complexity by ensuring that all segments are viable hosts for the features assigned, the broader issue of host compatibility and feature interaction remains a potential factor shaping the learnability of non-local segmental patterns and warrants further investigation. In sum, these distinct layers suggest that the advantage of tonal patterns is deeply rooted in differences in how tonal and segmental systems are represented and deployed in phonology.

5.3 Conclusions

To conclude, this study provides the first experimental investigation of the typological tone-segment asymmetry from a learnability perspective. The results show that the learning profiles of tonal and segmental counting rules closely mirror the typological distribution: segmental counting patterns, which are typologically rare, are consistently more difficult to learn. This learning asymmetry holds across different participant populations and generalizes across both consonantal and vowel features, suggesting that it reflects a robust and systematic bias.

Our findings argue against a purely phonetic account of this asymmetry. English speakers had difficulty distinguishing tones but still more readily learned tonal counting rules. The behavior of vowel rules in Experiment 2 demonstrates that how a given contrast is used in other tasks does not predict whether people will be able to acquire a counting rule based on that contrast. Vowels patterned with tones in the perceptual task and for learning the non-counting rules but patterned with tones when counting was required. Thus, the relevant distinction for counting rules appears to be the distinction between segments and tones. We propose that tonal features, by virtue of their autosegmental, prosodic, and melodic properties, are better suited for long-distance, non-local phonological processes such as the counting process that is approached here. More broadly, this study contributes to a growing body of work showing that phonological typology may be relevant to learnability biases. Future research may further investigate the nature of the tone-segment asymmetry across other feature types and phonological processes.

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